

Assignment 03: Advanced NLP Tasks with Hugging Face Pipelines

As we discussed in class, Hugging Face has created a comprehensive ecosystem of user-friendly open source tools for NLP, vision, and more. The core of this ecosystem is the Transformers library, which enables easy downloading of pretrained models along with their tokenizers, and allows for fine-tuning on custom datasets if required. Additionally, the library is compatible with TensorFlow, PyTorch, and JAX (via the Flax library).

The goal of this assignment is to deepen your understanding of Hugging Face pipelines, explore custom models, and demonstrate how to use these pipelines for various advanced NLP tasks including **text classification, question answering, and text generation**.

Tasks:

1. Write a Python script to demonstrate the basic usage of a pipeline for text classification. Use the *pipeline* function to create a sentiment analysis pipeline and classify a few sample texts.
 - Train a custom (pre-trained) text classification model using the Hugging Face library. Use a dataset such as **IMDb** for sentiment analysis.
 - Fine-tune the model and save it.
 - Use the trained model with the Hugging Face *pipeline* function to classify texts.
 - Print classification results.
2. Write a short Python script to demonstrate the basic usage of a pipeline for question-answering NLP tasks. Use the *pipeline* function to create a custom question-answering model.
 - Train a custom (pre-trained) question-answering model using the Hugging Face library. Use Stanford Question Answering Dataset **SQuAD**.
 - Fine-tune the model and save it.
 - Use the trained model with the *pipeline* function to answer questions based on a given context.
 - Print results.

3. Write a short Python script to demonstrate the basic usage of a pipeline for **text generation**. Use the *pipeline* function to create a text generation pipeline with a custom model.
- Fine-tune a text generation model such as **GPT-2** on a custom dataset (e.g., a collection of stories, articles, or other text).
 - Use the fine-tuned model with the *pipeline* function to generate text based on a given prompt.
 - Print text-generation results.

Deliverables:

- Neat and high-quality report with great details on steps, structure, design, selection of hyperparameters and results. [**Poor submissions that have snapshots of the code without any remarks/comments will be immediately discarded.**]
- The report should be submitted in PDF format starting with a cover page.
- Code File.

Grading Criteria:

- Correctness and completeness of the implemented tasks.
- Clarity and organization of the report.
- Proper use of Hugging Face pipelines and pre-trained/custom models.
- Quality of the discussion and observations in the report.

References:

- Hugging Face Pipeline Documentation: https://huggingface.co/transformers/main_classes/pipelines.html
- Fine-tuning a model: <https://huggingface.co/transformers/training.html>
- Hugging Face Documentation: [Hugging Face - Documentation](#)
- GPT-2: <https://huggingface.co/openai-community/gpt2>
- SQuAD dataset: <https://huggingface.co/datasets/rajpurkar/squad>

Due date: July 28.